

LIMITS & CONTINUITYContinuity at $x = a$:

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Standard limits:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

L'Hôpital's Rule:

If $f/g \rightarrow \frac{0}{0}$ or $\frac{\infty}{\infty}$:

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

Rational fcn at infinity (degree rule):

top < bot $\rightarrow 0$, top = bot $\rightarrow$ coef ratio, top > bot $\rightarrow \pm \infty$ **COMMON DERIVATIVES**

$$\frac{d}{dx} [\sin x] = \cos x$$

$$\frac{d}{dx} [\cos x] = -\sin x$$

$$\frac{d}{dx} [\tan x] = \sec^2 x$$

$$\frac{d}{dx} [\sec x] = \sec x \tan x$$

$$\frac{d}{dx} [\csc x] = -\csc x \cot x$$

$$\frac{d}{dx} [\cot x] = -\csc^2 x$$

$$\frac{d}{dx} [e^x] = e^x$$

$$\frac{d}{dx} [\ln x] = \frac{1}{x}$$

$$\frac{d}{dx} [a^x] = a^x \ln a$$

$$\frac{d}{dx} [\log_a x] = \frac{1}{x \ln a}$$

$$\frac{d}{dx} [\arcsin x] = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} [\arctan x] = \frac{1}{1+x^2}$$

IMPLICIT DIFFERENTIATION

1. Differentiate both sides w.r.t. x .
2. Each y -term gets a dy/dx (chain rule).
3. Group all dy/dx on one side.
4. Factor and solve for dy/dx .

DERIVATIVE RULES

Power $\frac{d}{dx} [x^n] = n x^{n-1}$

Constant $\frac{d}{dx} [c] = 0$

Multiple $\frac{d}{dx} [cf] = c f'$

Sum $\frac{d}{dx} [f \pm g] = f' \pm g'$

Product $\frac{d}{dx} [fg] = f'g + fg'$

Quotient $\frac{d}{dx} \left[\frac{f}{g} \right] = \frac{f'g - fg'}{g^2}$

Chain $\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$

EXTREMA & TESTSCritical pt: $f'(c) = 0$ or undefined.1st-deriv test (sign change of f' at c): $+$ $\rightarrow$ $-$: local max $-$ $\rightarrow$ $+$: local min2nd-deriv test (if $f'(c) = 0$): $f''(c) > 0$: local min $f''(c) < 0$: local max

FUNDAMENTAL THEOREM

Part 1 — Derivative of an integral:

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Part 1 with chain rule (the trap):

$$\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x)) \cdot g'(x)$$

Part 2 — Evaluating a definite integral:

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f

u-SUBSTITUTION

1. Pick u = inner function.
2. Compute $du = u'(x) dx$.
3. Rewrite the integral entirely in u .
4. For definite integrals, change bounds to u -values.

PARTICLE MOTIONVelocity: $v(t) = s'(t)$ Acceleration: $a(t) = v'(t)$ Displacement on $[a, b]$:

$$\int_a^b v(t) dt$$

Total distance traveled on $[a, b]$:

$$\int_a^b |v(t)| dt$$

Speeding up $\Leftrightarrow$ **v and a have the same sign.****INTEGRATION RULES**

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

APPLICATIONSAverage value of f on $[a, b]$:

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$

Area between two curves on $[a, b]$:

$$A = \int_a^b [\text{top} - \text{bottom}] dx$$

MEAN VALUE THEOREM

If f is continuous on $[a, b]$ and differentiable on (a, b) , there exists c with:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Rolle's Theorem (special case):

If $f(a) = f(b)$, there exists c with $f'(c) = 0$.